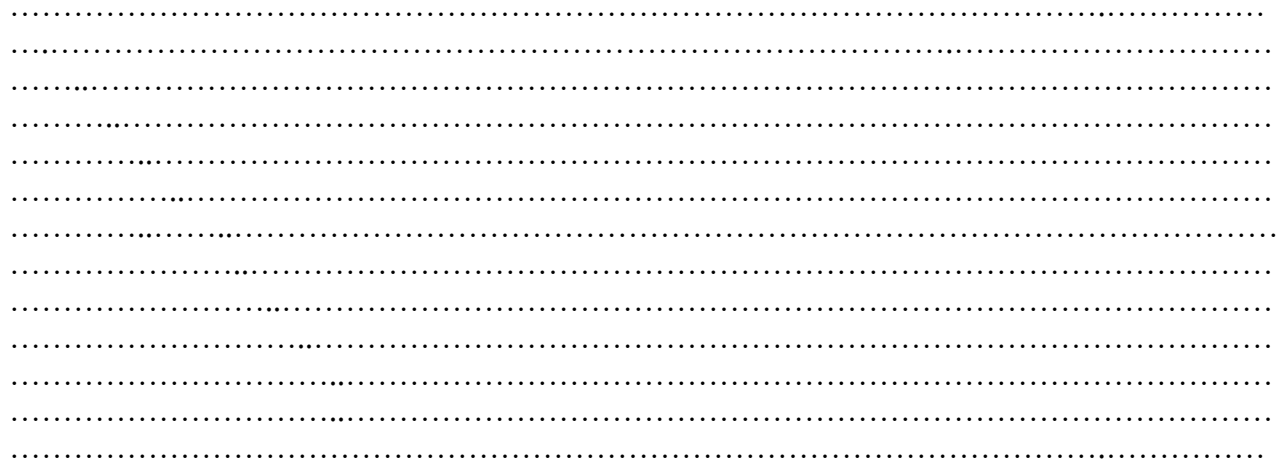


a) The policy makers of an economic system described by a standard IS-LM model decide to raise, at the same time and by the same amount, both government spending and net taxes so that $\Delta G = \Delta T > 0$. Show in the graph and explain how this decision will affect the equilibrium levels of output and of interest rate, denoting by Y_0 and i_0 the values of these variables at the initial equilibrium (call it '0' in the graph), and by Y_1 and i_1 those they will take on at the new equilibrium (to be denoted by '1' in the graph). Finally, discuss how and why private saving and investment will have changed in this new equilibrium.



- [illegible]

[illegible]

Question 2

True or False?

Explain whether the following statements are true or false. Motivate your answer in a brief but rigorous way. Making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

- a) “In a closed economy where public saving is negative, and equal to -50 billion Euros, and investment is equal to 50 billion Euros, private saving must be equal to zero in equilibrium”.

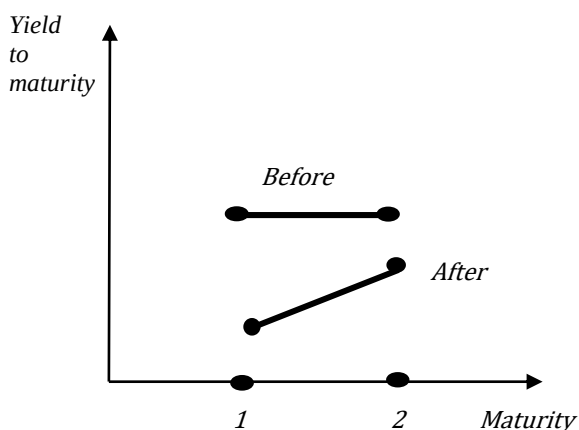
[illegible]

- b) “In a closed economy where only the goods market exists, government purchases of goods and services $\bar{G}$ and net taxes $\bar{T}$ are exogenous, private saving is an increasing function of disposable income and the investment function is $I = \bar{I} + d_1 Y$, where the parameter d_1 is greater than zero, a decrease in autonomous consumption c_0 leads to a new equilibrium where private saving is necessarily lower”. (Hint: you are NOT being asked to derive the expression of the change in private saving, but only to explain what the sign of this change is going to be, and why).

[illegible]

Question 3

- a) In country Gamma, prices are constant and people expect the prices to be constant in the future as well. In the economy, there are only one-year bonds and two-year bonds. In period $t + 1$ ('the future') consumption demand depends only on disposable income at $t + 1$ and investment demand depends on the values of the interest rate and income at time $t + 1$. At time t ('present'), consumption is increasing in total wealth that is the sum of human, financial and housing wealth. Investment is exogenous, $I = \bar{I}$, money demand has the usual functional form and M and G are exogenous. At time t , the price of houses decreases and this decrease is believed to be transitory ('it is believed that the decrease is limited to the current period') by households. Suppose that initially the yield curve was flat. Draw the yield curve for country Gamma at time t in the graph below and explain how the position and slope will change due to the decrease in the house prices.



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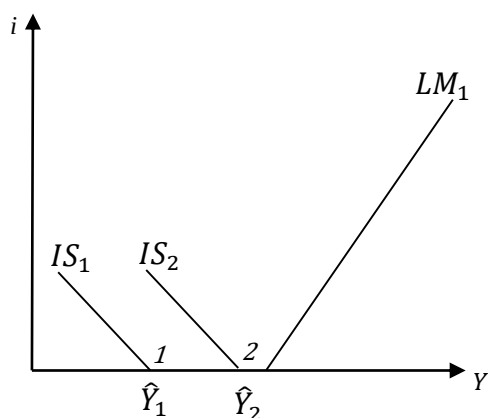
- b) The government of country Beta announces a permanent increase in taxes. At the same time, the central bank announces that it will implement policies such that the reserve coefficient θ will decrease permanently. In the stock market, will the price of stocks increase or decrease? Explain by referring to the two points below (analyze each case separately):
- the announcements were not expected by households;
 - the announcements were expected by households, so that the announcements just confirm the expectations that the market already had.
- (Hint: When answering, assume that the economy can be described by a non-standard IS-LM model).

.....

[illegible]

a) Alpha is a closed economy described by a non-standard IS-LM model where investment depends on real interest rate (r), rather than on the nominal interest rate (i).

- Give the definition of the real interest rate and write the relationship between the real interest rate (r) and the nominal interest rate (i).
- Suppose that the economy is in the equilibrium such as point '1' in the following graph. Initially, the economy is in a liquidity trap with equilibrium income equal to $\hat{Y}_1$. Economic authorities are not satisfied with $\hat{Y}_1$; current and expected future inflation rates are equal to zero. In order to increase the equilibrium level of income, economist Giky thinks that it could be a good idea to generate expectations of future positive inflation (and therefore higher than current inflation). Economist Michael, on the contrary, thinks that this not a good idea and suggests to generate expectations of decreasing prices. Who do you agree with? In the graph, show the impact of the opinion you agree with; indicate the new equilibrium as point '2' and explain through which channels it affects the income.

[illegible]

- b) Consider a country that, in period $t = 1$, inherits from the previous period $t = 0$, a debt-to-GDP ratio of 60%: $b_0 \equiv B_0/Y_0 = 0.6$. Moreover, the real interest rate and the rate of economic growth are constant and equal to 6% and to 0%, respectively ($r = 0.06, g = 0$). Finally, the ratio of the primary deficit to GDP is 3%. Suppose now that, at time $t = 2$, the government intends to stabilize debt-to-GDP ratio $b(\equiv B/Y)$ at the value observed at $t = 1$. In other words, government wants b to continue to take on the $t = 1$ value (b_1) in the periods $t = 2$ and in all subsequent periods. Compute the new value, in order to reach this objective, that primary deficit to GDP ratio should take on to stabilize b at time $t = 2$ and onwards. Comment.

[illegible]

Question 5

Consider an open economy operating under a flexible exchange rate regime. Domestic prices, P and foreign prices, P^* are both constant. In the pair of graphs ‘non-standard open economy IS-LM model’-‘interest parity condition’, show the initial equilibrium denoting it by ‘0’, and denoting by i_0, Y_0 and E_0 , the values of the interest rate, output and exchange rate at that equilibrium.

- a) Suppose that there is a fall in the foreign prices P^* , while domestic prices and all the other exogenous, both domestic and foreign, variables remain constant. (*Hint: you are only asked to consider a one-time variation in foreign prices. Once the foreign prices have fallen, they keep on being equal to new low level and the economy is described by a fixed price model*). In the non-standard IS-LM model, call '1' the new equilibrium that will be reached. Explain the effects of this exogenous decrease in P^* on the equilibrium levels of the interest rate, output and exchange rate of the economy we are considering.

[illegible]

- b) What would have been the impact on the equilibrium level of income of a decrease in foreign prices P^* if the country had had a fixed exchange rate regime and the central bank had intervened to keep the exchange at its initial level, E_0 ? Show in the same graph used in point a) the new equilibrium that will be reached in this case and denote it by '2'. Explain the reasons of the differences in income observed under flexible exchange rate and fixed exchange rate regimes.

[illegible]

Question 6

True or False?

Explain whether the following statements are true or false. Motivate your answer in a brief but rigorous way, by making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

- a. In an open economy with a flexible exchange rate and where the domestic nominal interest rate is at its *zero lower bound* ($i=0$), for a nominal exchange rate E greater than $E^e/(1+i^*)$ – where, as usual, E^e and i^* are taken as given – no one would be willing to hold foreign bonds – investors would be willing to hold domestic bonds only. [Hint: assume risk neutrality – investors make their decisions by taking into account the expected returns on domestic and foreign bonds only.]

[illegible]

- b) A government that inherits from the previous period ($t - 1$) a stock of government debt greater than zero, that is running primary surplus and that faces a real cost of borrowing smaller than the growth rate of the economy ($r < g$), can stabilize at time t the debt-to-GDP ratio by raising the primary surplus-to-GDP ratio by an appropriate amount, and/or implementing policies leading to an increase in g .

[illegible]

[illegible]